Chanchal Roy

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Professional Summary

MCA student and Python developer specializing in Agentic AI, Large Language Models (LLMs), FastAPI, LangGraph, and Retrieval-Augmented Generation (RAG). Builds and deploys AI-powered backend applications using Python, Docker, MongoDB, and Google Cloud Platform (GCP). Strong foundation in REST API design, prompt engineering, and cloud-native, scalable system architecture.

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, SQL
- **AI / Machine Learning:** Agentic AI, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Gemini API, LLM Integration
- **Frameworks & Libraries:** FastAPI, React.js, Tailwind CSS, REST APIs
- **Databases:** MongoDB, MySQL
- **Cloud & DevOps:** Google Cloud Platform (GCP), Docker, GitHub Actions, Terraform, CI/CD
- **Tools:** Git, GitHub, VS Code, Postman, Figma

Projects

Health Insight – Agentic AI Health Assistant *Python, LangGraph, FastAPI, Gemini API*

- Architected a multi-agent workflow with LangGraph to automate health-data analysis and generate personalized recommendations.
- Integrated the Gemini API with a FastAPI backend, reducing manual analysis effort and improving response accuracy.
- Designed modular agent nodes for data ingestion, reasoning, and recommendation generation.

AI Chatbot Platform *Python, FastAPI, REST APIs*

- Built a conversational AI chatbot backend using FastAPI, applying structured prompt engineering to improve response quality.
- Designed and documented REST API endpoints for scalable client integration.
- Implemented a modular backend architecture to support future feature expansion.

Personal Portfolio Website *React.js, Tailwind CSS*

- Designed and developed a responsive portfolio site to showcase AI and full-stack projects.
- Optimized page load performance and accessibility using semantic HTML and Tailwind CSS.

Experience

Independent AI Developer – Personal Projects 2024 – Present

- Designed and deployed multiple AI-powered applications using Docker, GitHub Actions, FastAPI, and Google Cloud Platform.
- Developed and tested REST APIs and backend services, applying CI/CD pipelines for automated builds and deployment.
- Collaborated on open-source repositories, contributing code reviews and feature implementations via GitHub.

Education

Master of Computer Applications (MCA) 2024 – Present
Pondicherry University

Bachelor of Computer Applications (BCA) 2021 – 2024
Maulana Abul Kalam Azad University of Technology – CGPA: 7.34/10

Certifications

- Google Cloud Platform Fundamentals – *In Progress*
- Agentic AI & Large Language Models – *In Progress*

Languages

English • Hindi • Bengali